

```
import pandas as pd
import matplotlib.pyplot as plt

# Load Dataset
df = pd.read_csv('/content/Mall_Customers.csv')

# First 5 rows
print(df.head())

# Dataset Information
print(df.info())

# Summary Statistics
print(df.describe())

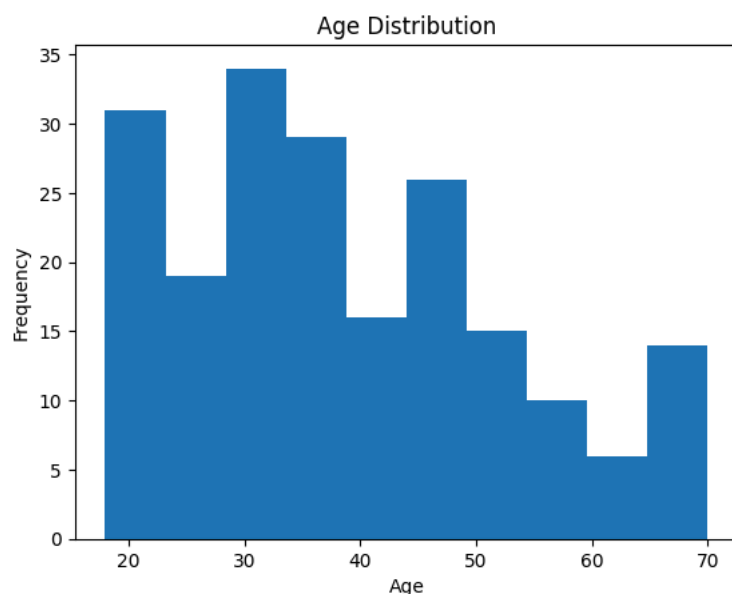
# Missing Values
print(df.isnull().sum())
```

```
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                15                39
1           2    Male   21                15                81
2           3  Female   20                16                 6
3           4  Female   23                16                77
4           5  Female   31                17                40
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   CustomerID            200 non-null   int64
 1   Gender                200 non-null   object
 2   Age                   200 non-null   int64
 3   Annual Income (k$)    200 non-null   int64
 4   Spending Score (1-100) 200 non-null   int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
None
```

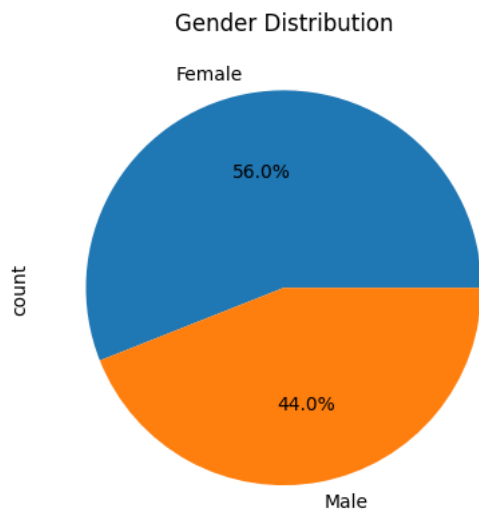
	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

```
CustomerID
0
Gender
0
Age
0
Annual Income (k$)
0
Spending Score (1-100)
0
dtype: int64
```

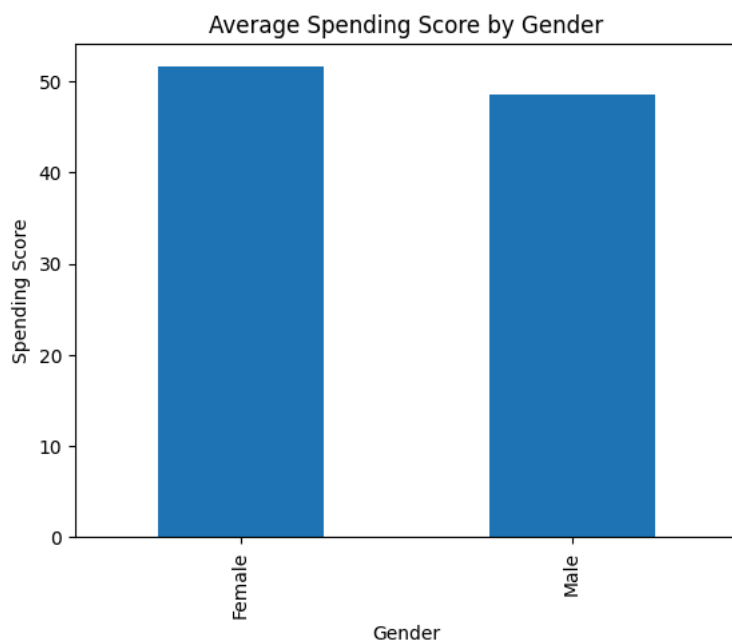
```
df['Age'].plot(kind='hist')
plt.title('Age Distribution')
plt.xlabel('Age')
plt.show()
```



```
df['Gender'].value_counts().plot(kind='pie', autopct='%1.1f%%')  
plt.title('Gender Distribution')  
plt.show()
```



```
df.groupby('Gender')['Spending Score (1-100)'].mean().plot(kind='bar')  
plt.title('Average Spending Score by Gender')  
plt.ylabel('Spending Score')  
plt.show()
```



Limitations

- The dataset size was limited and may not represent all customer groups.
- Some customer behavior factors were not available in the dataset.
- The analysis was based only on available historical data.

Recommendations

- Businesses should target young customers through personalized marketing campaigns.
- Customer engagement strategies should focus on high-spending customer groups.
- Additional customer behavior data should be collected for deeper analysis and better prediction.

